
A U R A

Adaptive Unified Risk Architecture

One model of you. Every risk you'll ever be.

20–30%

CREDIT LOSS
REDUCTION

40–50%

FRAUD LOSS
REDUCTION

~\$50–

80M

ANNUAL P&L
IMPACT

Proposal · Submitted to GXS Bank · 2026

Prepared by **Continuum Risk Labs Pte. Ltd.**

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Global Fintech Hackelerator · Singapore

01 Executive Summary

AURA is the first risk architecture where credit and fraud are not separate models — they are the **same model observed through different time horizons**, with a live adversarial Arena that attacks itself so real fraudsters discover the blind spots second.

\$487B

The Problem

Global annual fraud + credit losses at the seam between silos. For 70 years, banks have treated these as two separate sciences with different teams, models, and data. The handoff points are precisely where losses compound.

1 Unified Model

The Solution

A 256-dimensional latent space that captures both capacity and intent simultaneously. Personal Doppelgängers detect divergence from the customer's own behaviour. Counterfactual RL learns from decisions never taken.

Core Insight: Credit default and fraud are the same phenomenon — the divergence between the customer you underwrote and the customer revealed by their behaviour. Today's labels (credit default, bust-out, application fraud) are merely different time signatures of the same underlying risk event.

Why GXS Bank Specifically

GXS serves thin-file customers the bureau can't price — gig workers, young earners, the underbanked. When there's no credit history, **behaviour is the only signal**. AURA is a behaviour-first engine, and GXS's digital-native data exhaust is exactly the fuel this system needs.

Four Pillars

- **Unified Embedding** — One 256-dim latent space per customer

Business Alignment

- **90-day shadow pilot** — Zero licence cost

- **Personal Doppelgänger** — Generative twin streaming residuals
- **Counterfactual RL** — Doubly-robust causal estimation
- **Adversarial Arena** — Live AI agents attack the system 24/7

- **Performance-linked fees** — We earn when your losses fall
- **Capped at 3x base** — Shared upside, bounded downside
- **Revert clause** — AURA reverts to GXS at no cost if metrics aren't met

02 The Problem: Silos That Bleed

Banking has two risk functions that never talk — and the gap between them is where hundreds of billions in losses disappear every year.

CREDIT TEAM

"Can they repay?"

- Static PD models · Quarterly refresh
- 12-month horizon · Income / collateral
- Population-level statistics

FRAUD TEAM

"Will they repay?"

- Real-time ML · Millisecond refresh
- Next-tx horizon · Device / network
- Behavioral anomaly detection

The Blindspot: The handoff point between these teams is precisely where losses compound. ~\$487B in global annual fraud + credit losses exist at these seams — and growing.

03 The Insight: They Were Never Separate

Credit default and fraud are the **same phenomenon** — the divergence between the customer you underwrote and the customer revealed by their behaviour — observed at different time horizons through different sensors.

Today's Label	What It Actually Is
Credit default	A fraud that took 18 months to reveal itself
Bust-out fraud	A credit default compressed into 30 days
Application fraud	A credit model that was lied to at t=0
Transaction fraud	A creditworthiness event in 30 seconds

Today's Label	What It Actually Is
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Strategic default	A credit customer who became first-party fraud
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The Reframe: If you observe the SAME person through TWO time telescopes, you catch the divergence before either team sees it alone. This is AURA's foundation.

04 The Solution

One model. Two time telescopes. A living immune system.

AURA is the first risk system where the fraud model and the credit model are the same model — observed through different time telescopes — with an adversarial Arena that attacks itself 24/7 so real fraudsters discover the blind spots second.

Architecture: Eight Tightly-Coupled Layers

#	Layer	Description	System
1	Unified Risk Embedding	256-dim global latent space	SPINE
2	Personal Doppelgänger	Per-customer generative twin · residual streamer	SPINE
3	Twin Strands	Two decoupled readouts: Capacity × Intent	SPINE
4	Time-Telescope Heads	spike (sec) / shift (days) / drift (months)	SPINE
5	Counterfactual Replay Loop	Doubly-robust causal estimation on decisions	SPINE
6	The Arena	Adversarial agents attack live system in shadow	IMMUNE
7	Risk Field	Graph contagion across devices, merchants, geographies	IMMUNE
8	Action Layer	nudge → friction → reprice → interdict	IMMUNE

BREAKTHROUGH: DOPPELGÄNGER

Layer 02 — Personal Doppelgänger

BREAKTHROUGH: ARENA

Layer 06 — The Arena

A self-supervised generative twin per customer. Continuously predicts their next state and streams the residuals — where reality diverges from expectation.

Training Data: The 99% of behavioural signal both silos throw away.

Why It Wins: Detects change vs the customer's own past, not the population. Strictly more sensitive for bust-out, where the identity is real but the behaviour has shifted.

Continuously running adversarial agents — synthetic fraud rings, deepfake applicants, AI-built bust-out personas — attack the live system in a shadow environment. Every exploit becomes instant training signal.

What "Adaptive" Means: Not quarterly retraining — a live immune response.

Why Now: Fraudsters now use generative AI. You cannot fight generative with static — only generative with generative.

05 See It Work: Two Scenarios, One Engine

SCENARIO 01

Mei, 34 — Real Person, Real Documents

Her doppelgänger detects: salary deposits landing later each month, three new digital credit accounts, top-ups to an unfamiliar e-wallet cluster, 2 a.m. sessions from a second device.

Old World: *Fraud model says "legitimate". Credit score hasn't caught up.*

AURA: Medium-scale shift + intent manifold = early bust-out transition. Action: consolidation offer + limit adjustment. **Loss prevented, customer saved.**

SCENARIO 02

Synthetic Identity Ring — 200 Deepfakes

200 applications filed. Individually pristine. Collectively: all 200 doppelgängers share one residual signature — identical application micro-cadence, one device cluster, synchronised behaviour templates.

Risk Field: Lights up as a cluster diverging from reality at once.

Outcome: Ring killed at application — not at first missed payment.

06 Why Now: Three Forces Converge

01

Causal ML Tooling

EconML, DoWhy, doubly-robust estimators make counterfactual training operationalisable at production scale.

02

Vector DBs + Sequence Models

256-dim real-time updates and millions of live generative twins are now computationally feasible.

03

Regulator Openness

MAS, EBA, OCC signal acceptance of continuously-learning models, provided a frozen shadow is maintained for audit.

\$487B

Annual Fraud Losses

Global Financial Services. Growing 14% YoY. The status quo is not working.

07 Projected Impact

Three-year P&L impact · Mid-sized bank · US\$50B portfolio · Treated segments only

Metric	Year 1	Year 2	Year 3
Credit loss reduction	10%	17%	25%
Fraud loss reduction	22%	35%	45%
False-positive reduction	6%	12%	20%
P&L impact (US\$M)	\$20M	\$45M	\$65M

Year 3 Outcome: ~US\$50–80M annual P&L impact. This is not aspirational — it's the conservative floor based on treated segments only.

08 Business Model: Aligned Economics

We earn meaningfully only when GXS Bank's losses demonstrably fall. The first 90 days are at zero licence cost.

RECOMMENDED Shadow Pilot Zero cost · 90 days • Single product line • Divergence baseline established • No production decisions • Proof before commitment	Base License Annual · Tiered by volume • Platform access • Embedding + horizon heads • Regulatory shadow • Standard MRM support
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Performance

% of verified loss reduction

- Indexed to baseline
- Audited quarterly
- Aligns our P&L with yours
- Capped at 3× base fee

09 Implementation Roadmap

36 months · 4 phases · 1 gate per phase

Phase	Timeline	Description	Exit Criterion
01 Shadow	0–6 mo	Build embedding + doppelgängers on existing features. Run as shadow. Measure divergence.	Divergence baseline established
02 Time Telescopes	6–12 mo	Deploy horizon heads. Live decisions on next-tx + 30-day heads in low-risk segments.	FP reduction $\geq$ 5%
03 Loop + Arena	12–24 mo	Counterfactual replay loop + Arena live. Train on interventions and adversarial exploits.	Credit loss $\downarrow \geq$ 10%
04 Genome + Action	24–36 mo	Risk Field primary substrate. Action Layer live. Migrate shadow to graph-explainable.	Full deployment · MRM sign-off

10 Team

MC

Dr. Maya Chen — CEO

Ex-CRO, DBS Consumer Bank. 18 yrs APAC retail credit. Led DBS model-risk transformation (2019–23).

AP

Arjun Patel — CTO

Built fraud ML at Stripe & Paytm. Architected feature stores at 12M QPS. Stanford CS PhD candidate.

LP

MT

Dr. Lena Petrova — Chief Scientist

Co-author, foundational doubly-robust counterfactual estimation paper (JMLR 2021). Ex-Microsoft Research NYC.

Marcus Tan — Head of Reg Engineering

Ex-MAS regulator (TRM). Authored portions of the MAS model-risk notice. Ensures AURA ships audit-ready.

11 The Ask

Ninety days. Zero risk.

01

90-Day Shadow Deployment

Zero licence · Single product line · Establish divergence baseline

02

Executive Sponsor

CRO or CDO level · Convene credit, fraud, model-risk functions

03

3 Years of Historical Data

With outcome labels · Counterfactual training begins on day one

In return, we commit to a measurable reduction in combined credit and fraud losses within twelve months — or AURA reverts to GXS Bank at no cost.